

Assignment Title	Individual Assignment
Learning outcomes covered	LO1 – LO5
Assignment Mode	Take home Assignment
Maximum Marks	100
Contribution to the Final Grade	25%
Date published	7 th Week
Deadline for submissions	23 rd of September 2026
Mode of Submission	Zip File with Report and template file

Assignment Description: In this assignment, students will understand a real-world business scenario and design a cloud solution using AWS services. Students must create an architectural design and implement the infrastructure using relevant cloud services.

Refer to the case study given below and follow the tasks.

Prepare a proper documentation including the necessary information mentioned in each task.

Tomorrow Day, a division of News Corp., is a global leader in news and business information services. With several renowned brands, the company serves millions of readers worldwide through digital, mobile, and print platforms. As its digital audience in Asia continued to grow, *Tomorrow Day* sought a more efficient and scalable infrastructure solution to support business expansion and improve customer experience.

Tomorrow Day faced a critical infrastructure challenge when the lease for its China data center was nearing expiration. At the same time, ABC.com was serving approximately 12.8 million users in Asia and generating around 90 million page views each month. The company needed a solution that could reduce latency for Asian users, eliminate the complexities of managing physical infrastructure, and enable faster deployment of new products and services. IT teams were spending significant time maintaining servers and data centers instead of focusing on innovation and revenue-generating activities.

Assume you are the cloud architect, who is assigned to provide a cloud based solution to address above mentioned challenges of *Tomorrow Day*.

Task 1 – Read the given case study and identify,

- Business requirements
- Functional requirements
- Expected users of the system

Task 2 – Design an architectural diagram stating the cloud solution that you proposed for the above mentioned business scenario.

Your diagram should include:

- Network layer (at least one Virtual Network with public and private subnets)
- Compute layer (at least one Instance)
- Storage layer (at least one)
- Database layer (at least one RDS or EC2 Database server)

You may use draw.io or Lucid charts to draw the architecture diagram. Use relevant icons while drawing the diagram.

Task 3 – Implement your proposed cloud solution in the cloud platform. Take screenshots of *Details* tab of every cloud resource (ex:- EC2 instances, VPC, Security groups, etc ...) that you provision and configure.

Always name of your resource contains your Student ID with the resource name.

Ex:- <ITXXXXXX_VPC>

Task 4 – Create an Infrastructure as Code (IaC) template which provisions all the cloud resources that you configured in the previous step.

You need to submit:

- CloudFormation YAML/JSON template
- Screenshots of successful stack deployment
- Screenshots of created resources



Task 5 – Perform the following tests and validate them with the relevant screenshots.

- Verify web server accessibility from the internet
- Verify database accessibility only from the web server
- Verify instances are deployed in correct subnets
- Verify security group rules.

Important

- Use AWS Academy Sandbox Environment to complete the implementation.
- CloudFormation template must be written by you.
- Resources should be tagged appropriately.
- Include screenshots for all major deployment steps.
- Use a naming convention for resources. (Ex- ITXXXXX_Webserver / ITXXXXX_VPC).

Usage of AI: Students may use AI tools for pre-task activities including brainstorming, outlining, and initial research. The final submission must show how students have developed and refined these ideas independently.



Marking Scheme

Criteria	Excellent 85 - 100	Good 70 - 84	Satisfactory 55 - 69	Poor 40 - 54	Very Poor 40 - 0
Scenario Analysis and Requirements (10 marks)	Comprehensive analysis of scenario with all requirements accurately identified and justified	Good analysis with most requirements identified correctly.	Basic analysis with key requirements identified.	Limited analysis with several missing requirements.	Little or no analysis requirements mostly incorrect or missing.
Architecture Design (15 marks)	A well documented architecture with clear justification of components and services.	Appropriate architecture with minor weaknesses and adequate justification.	Functional architecture meeting basic requirements.	Incomplete architecture with significant design issues.	Poor or missing architecture design.
Implementation of proposed cloud solution using console (20 marks)	All services correctly implemented and configured; demonstrates advanced use of cloud platform features.	Most services implemented correctly with minor configuration issues.	Core services implemented and functional.	Partial implementation with several errors.	Minimal or no implementation completed.
IaC Development (25 marks)	Complete IaC solution, modular, reusable, well-documented, and error-free.	Functional IaC implementation with good structure and minor issues.	Basic IaC implementation covering essential resources.	Incomplete IaC with several errors or missing resources.	Little or no IaC development demonstrated.

Deployment Evidence and Testing (15 marks)	Comprehensive deployment evidence with detailed testing, validation, and results.	Good deployment evidence and adequate testing.	Basic deployment evidence with limited testing.	Insufficient deployment evidence or testing.	No deployment evidence or testing provided.
Creativity of the solution (5 marks)	Demonstrates exceptional innovation, optimization, and unique features beyond requirements.	Shows several creative enhancements.	Some creativity evident in implementation.	Limited creativity or originality.	No evidence of creativity.
Report (10 marks)	Well-structured, comprehensive documentation	Well-organized report with minor issues.	Adequate report covering key aspects.	Poorly structured report with missing details.	Report missing or of unacceptable quality.
Viva (15 marks)	Demonstrates excellent understanding of concepts and confidently answers all questions.	Good understanding with satisfactory responses to most questions.	Basic understanding demonstrated.	Limited understanding with difficulty answering questions.	Unable to explain work or answer questions satisfactorily.

Note – Deductions in marks may apply if any of the following conditions are observed.

- Late submissions
- Missing configurations
- Missing evidence (screenshots)
- Academic Integrity Penalties
 - Architecture copied without modification
 - CloudFormation copied from online repository without understanding
 - AI-generated template submitted without customization/documentation